ECE343 Spring 2025

Objective 7.1:

Biot Savart



UNITED STATES
AIR FORCE
ACADEMY

SCHEDULE

Lesson	Topics	Reading	Notes
24	BVP (Capacitor, Coax), Electric Potential	5.16	LO 5.3
25	Electric Boundary Conditions	5.18-5.22	LO 5.4
26	BC's/Capacitance	5.23-5.24	
27	Electrostatic Lab	Lab packet	
	Chapter 6: Steady Current & Conductivity		
28	Current density, Conductivity, Resistance	Ch. 6	
	Chapter 7: Magnetostatics		
29	Magnetic Fields, Gauss' Law for Magnetism	7.1-7.3	LO 7.1-7.2
30	Ampere's Law	7.4-7.5, 7.9	LO 7.3
31	Ampere's Law Applied	7.6-7.8	LO 7.3
32	Magnetic Boundary Conditions	7.10-7.14	LO 7.4
33	Magnetostatics Lab	Lab packet	
	Chapter 8: Time-Varying Fields		
34	Faraday's Law, Displacement Current, Maxwell Eqns	8.3, 8.8, 8.9	LO 8.1
35	Faraday's Law, Displacement Current, Maxwell Eqns	8.3, 8.8, 8.9	LO 8.2-8.3
	Chapter 9: Plane Waves in Lossless Media		
36	Maxwell's Equations in lossless, source-free region	9.1-9.2	LO 9.1
37	Maxwell's Equations in lossless, source-free region	9.1-9.2	LO 9.1
38	Uniform Plane Waves & Polarization	9.4-9.6	LO 9.2-9.3
39	Normal Incidence	Vol 2, 5.6-5.7	LO 9.4-9.5
40	Oblique Incidence	Vol 2, 5.6-5.7	LO 9.5
41	Oblique Incidence	Vol 2, 5.6-5.7	LO 9.5
42	Incidence Lab	TBD	LO 9.5

Ch 7 Learning Objectives

7.1 I can use the Biot-Savart law to simple magnetostatic systems to find the magnetic field at an observation point.

7.2 I can articulate the meaning of both the integral form and point form of Gauss' Law for magnetism.

7.3 I can use Ampere's Law to calculate the magnetic field at an observation point for simple magnetostatic systems.

7.4 I can apply the magnetic field boundary conditions to calculate magnetic fields on both sides of a material boundary.

Similarities to Electrostatics

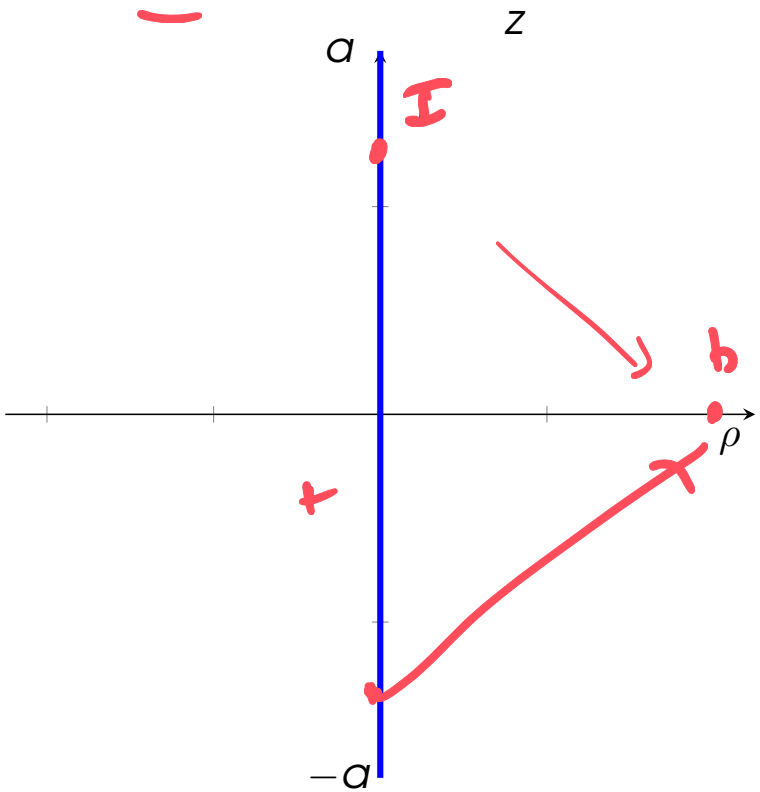
Electrostatics	Magnetostatics
$\vec{E}$ (V/m)	$\vec{H}$ (A/m)
$\vec{D}$ (C/m^2)	$\vec{B}$ (Wb/m^2) T
$\vec{D} = \varepsilon \vec{E}$	$\vec{B} = \mu \vec{H}$
$\vec{E} = \int \frac{dQ}{4\pi\varepsilon \lvert R \rvert^3} \vec{R}$	$\vec{H} = \int \frac{I \vec{d\ell} \times \vec{R}}{4\pi \lvert R \rvert^3}$
$\iint \vec{E} \cdot \vec{dS} = \frac{Q_{encls}}{\varepsilon}$	$\iint \vec{B} \cdot \vec{dS} = Q_{m,encl.} = 0$

Coulomb → Biot-Savart

Gauss

Biot-Savart: Line Charge (LC)

Find $\vec{H}$ at a distance b from the LC.



1. Draw a picture
2. Biot-Savart: $\vec{H} = \int \frac{I d\vec{\ell} \times \vec{R}}{4\pi |R|^3}$
3. $\vec{R} = \hat{\rho} b \pm \hat{z} z$

   $|R| = \sqrt{b^2 + z^2}$
4. $d\vec{\ell} = \hat{z}\, dz$
5. $d\vec{\ell} \times \vec{R} =$

$$\begin{bmatrix} \hat{\rho} & \hat{\phi} & \hat{z} \\ 0 & 0 & dz \\ b & 0 & z \end{bmatrix}$$

$\hat{\rho}(0-0) - \hat{\phi}(0 - b\,dz) + \hat{z}(0)$

$\hat{\phi}\, b\, dz$

Biot Savart: Line Charge

6 Integrate!

$$\vec{H} = \int_{-a}^{a} \frac{I\,(\hat{\phi}\,b\,dz)}{4\pi\,(b^2+z^2)^{3/2}}$$

$$= \hat{\phi}\frac{Ib}{4\pi}\int_{-a}^{a} \frac{dz}{(b^2+z^2)^{3/2}}$$

$$= \hat{\phi}\frac{Ib}{4\pi}\left[\frac{z}{b^2(b^2+z^2)^{1/2}}\right]_{-a}^{a}$$

$$= \hat{\phi}\frac{Ib}{4\pi}\left[\frac{a}{b^2(b^2+a^2)^{1/2}} - \frac{-a}{b^2(b^2+a^2)^{1/2}}\right]$$

$$\vec{H} = \hat{\phi}\frac{I\,a}{2\pi b\,(b^2+a^2)^{1/2}} \quad \text{"Finite"}$$

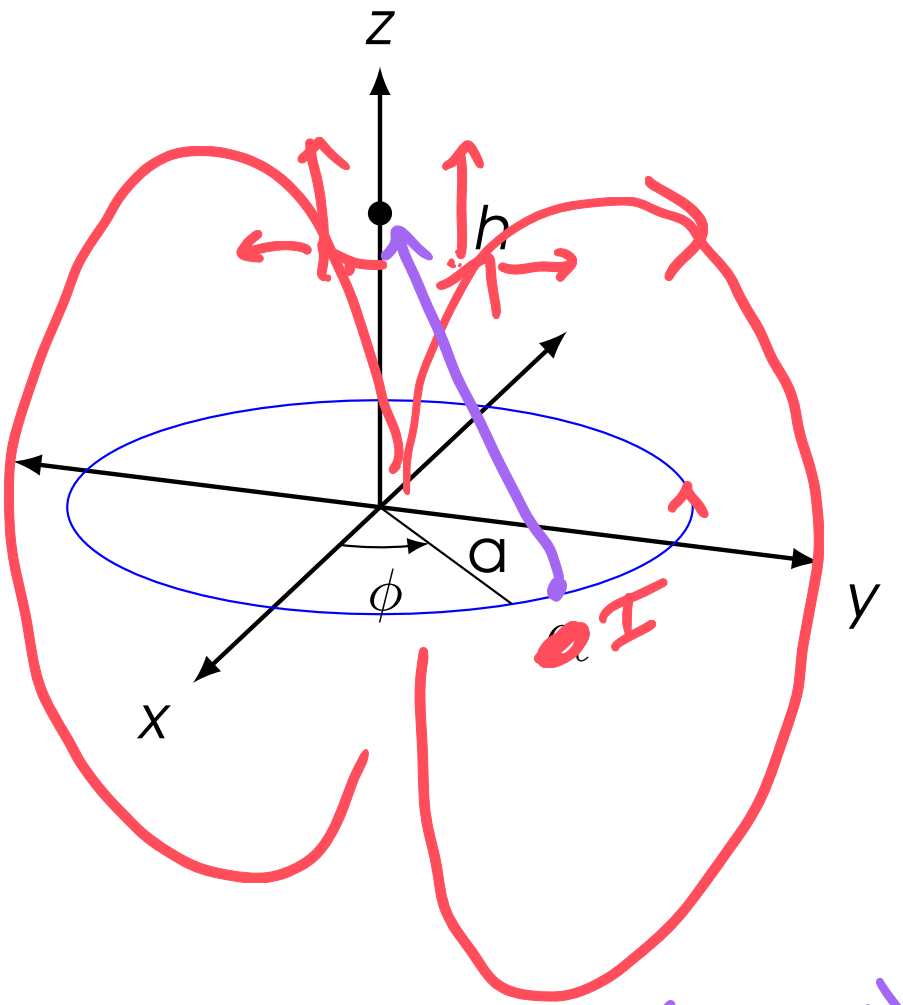
for "infinite" wire $\Longrightarrow$ $a >> b$

$$\vec{H} = \hat{\phi}\frac{I\,a}{2\pi b\,(a^2)^{1/2}}$$

$$\boxed{\vec{H} = \hat{\phi}\frac{I}{2\pi b}}$$

Biot-Savart: Wire Loop

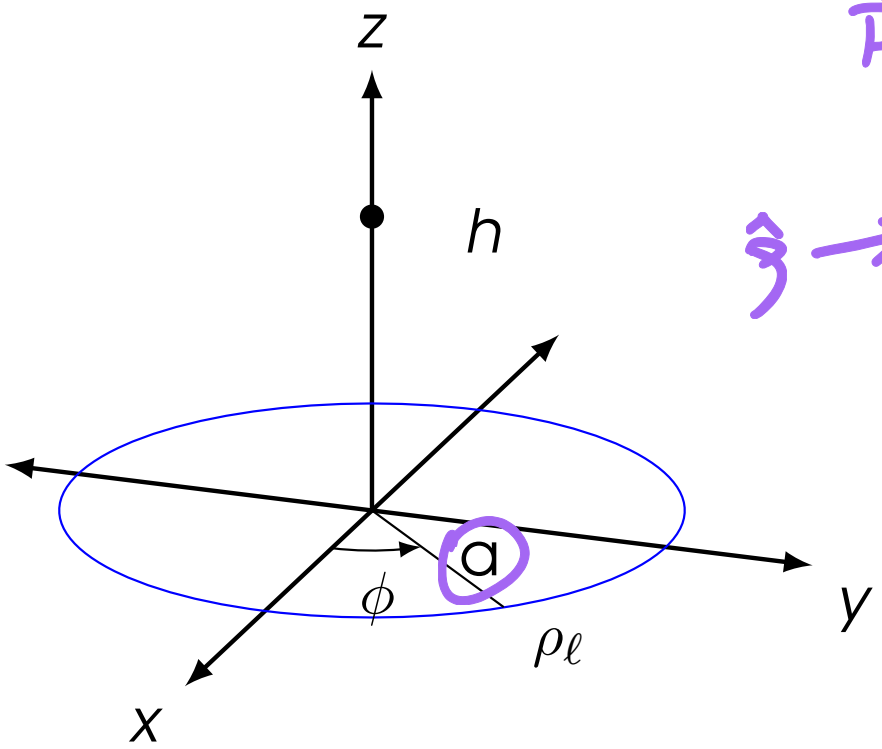
Find $\vec{H}$ at $z = h$.



1. Draw a picture
2. Biot-Savart: $\vec{H} = \int \frac{I d\vec{\ell} \times \vec{R}}{4\pi |R|^3}$
3. $\vec{R} = -\hat{y}a + \hat{z}h$
4. $|R| = \sqrt{a^2 + h^2}$
5. $d\vec{\ell} = \hat{\phi}\, a\, d\phi$
6. $d\vec{\ell} \times \vec{R} = \begin{vmatrix} \hat{x} & \hat{\phi} & \hat{z} \\ 0 & a\,d\phi & 0 \\ 0 & h & 0 \end{vmatrix}$

$\hat{x}(h\,a\,d\phi) - \hat{\phi}(0) + \hat{z}(a\,d\phi)(-a)$

BIOT-SAVART: WIRE LOOP



$\rho = a$

Integrate!

$$\vec{H} = \frac{I}{4\pi}\int_0^{2\pi} \frac{\hat{\rho}\, h a\, d\phi + \hat{z}\, a^2 d\phi}{(a^2+h^2)^{3/2}}$$

$$\hat{\rho} \to \frac{ha}{(a^2+h^2)^{3/2}}\int_0^{2\pi} \hat{\rho}\, d\phi \qquad \hat{\rho} = \hat{x}\cos\phi + \hat{y}\sin\phi$$

$$= 0$$

$$\hat{z} \to \frac{\hat{z}\, a^2}{(a^2+h^2)^{3/2}}\int_0^{2\pi} d\phi$$

$$= \hat{z}\,\frac{a^2 2\pi}{(a^2+h^2)^{3/2}}$$

$$\vec{H} = \hat{z}\,\frac{I a^2}{2(a^2+h^2)^{3/2}}$$